

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering ASSESSMENT TOOL 3 – NUMERICAL HACKATHON

Course Code:	ITA03	Course Title:	Mobile computing
CO Assessed:	CO3 – Precision (BL3)	Assessment:	Assessment Tool 3 – HACKATHON
Weightage:	30 % of CIA	Total Marks:	100
CO3 Statement:	Implement wireless networking and Mobile IP concepts for routing in cellular and ad hoc networks.. (BTL 3)		
Student Name:	K. Sujith Sai Vighnesh	Reg. No.:	192311199
Section / Batch:	5bt-A / 2023	Date:	12-08-2026

Code of Conduct

I K. Sujith Sai Vighnesh 192311199 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: K. Sujith Sai Vighnesh

Instructions

- Duration: 60 minutes
- Number of questions : 10

Faculty In-charge	Course Coordinator	Head of Department
Dr. Terrance Frederick	Dr. Terrance Frederick	
Signature: <u></u>	Signature: <u></u>	Signature: _____
Date: <u>12/8/26</u>	Date: <u>12/8/26</u>	Date: _____

Given

$$\text{Data Size} = 500 \text{ bytes}$$

$$\text{Tunnel Overhead} = 20 \text{ bytes}$$

$$\text{Total Packet size} = \text{Original packet} + \text{Tunnel header}$$

$$= 500 \text{ bytes} + 20 \text{ bytes}$$

$$= 520 \text{ bytes}$$

$$\text{Overhead Percentage} = \left[\frac{\text{Tunnel header size}}{\text{Original packet size}} \right] \times 100$$

$$= \frac{20}{500} \times 100 = 4\%$$

∴ The additional overhead for 500 byte data packet travelling through the Mobile IP is 20 bytes creating relative overhead of 4% $\approx$ 20 bytes

2)

Given

$$\text{Bandwidth} = 20 \text{ MHz} = 20 \times 10^6 \text{ Hz}$$

$$\text{Modulation} = 64\text{-QAM}$$

$$\text{Bits Per Symbol (n)} : 6 \text{ bits/symbol} \Rightarrow \log_2(64) = 6$$

$$\text{Theoretical Data Rate (R)} = \text{Bandwidth} \times \text{Bits Per Symbol}$$

$$= 20 \times 10^6 \times 6$$

3) Given

No of entries $[N] = 50$ entries

Size Per entry $S = 32$ bytes

Total Memory Required $[M_{total}] = \text{No of entries} \times \text{Size Per entry}$

$$= 50 \times 32 \text{ bytes}$$

$$M_{total} = 1600 \text{ bytes}$$

Convert to $[KB] \Rightarrow M_{total} = \frac{1600}{1024} \approx 1.5625 \text{ KB}$

Routing Table Requires 1600 bytes of memory

4) Given

Speed = 10 m/s

Time = 2 sec

$$\text{Distance} = \text{Speed} \times \text{Time}$$

$$= 10 \text{ m/s} \times 2 \text{ sec}$$

$$\text{distance} = 20 \text{ meters.}$$

5) Given

No of intermediate nodes $[N] = 5$ nodes

Initiating Source node = 1 Node

$$\text{Total RREQs} = \text{Source Generation} + \text{Intermediate Replication}$$

$$= 1 + N$$

$$= 1 + 5$$

$$= 6$$

6) Given

$$\text{Subnet Mask} = 255.255.255.240$$

binary of 240 $\Rightarrow$ 11110000

It leaves 4 bits for host address ($h=4$)

$$\begin{aligned}\text{Assignable Addresses} &= 2^h - 2 \\ &= 2^4 - 2 \\ &\Rightarrow 16 - 2 = 14,,\end{aligned}$$

Given

$$\text{Carrier freq} = 2.4 \text{ GHz}$$

$$\text{Channel Bandwidth (W)} = 20 \text{ MHz} = 20 \times 10^6 \text{ Hz}$$

$$\text{Symbol Rate (R}_s\text{)} = 1 \text{ MSPS} \Rightarrow 1 \times 10^6 \text{ symbols/sec}$$

$$\begin{aligned}\text{Spectral Efficiency} &= \frac{R_s}{W} \\ &= \frac{1 \times 10^6 \text{ symbol/sec}}{20 \times 10^6 \text{ Hz}} \\ &= \frac{1}{20} \Rightarrow 0.05 \text{ symbols/Hz}\end{aligned}$$

8) Given

$$\text{Ad hoc Nodes} = 50$$

$$\text{Range} = 250 \text{ m} = 0.25 \text{ km}$$

$$\text{Area} = 1 \text{ km}^2 \Rightarrow 10,00,000 \text{ m}^2$$

$$\text{Spatial Node Density (P)} \Rightarrow P = \frac{N}{A}$$

$$\Rightarrow \frac{50}{1} = 50 \text{ nodes/km}^2$$

$$\text{Connectivity Density} \Rightarrow D = P \times \pi R^2$$

$$= \frac{N \times \pi R^2}{A} =$$

$$D = 50 \times 0.19635 \text{ km}^2$$

$$D \approx 9.817$$

$$\Rightarrow \text{spacial Density} = 50 \text{ Nodes / km}^2$$

$$\Rightarrow \text{Connectivity Density} = 9.82 \text{ nodes per Transmission Area.}$$

9) Given

$$\text{Mobile Node Velocity (v)} = 30 \text{ m/s}$$

$$\text{Carrier frequency (f)} = 900 \text{ MHz} \Rightarrow 900 \times 10^6 \text{ Hz}$$

$$\text{Speed of light} = 3 \times 10^8 \text{ m/s}$$

$$\text{Wavelength } [\lambda] = c/f$$

$$= \frac{3 \times 10^8 \text{ m/s}}{900 \times 10^6 \text{ Hz}} = \frac{3}{9} \Rightarrow \frac{1}{3} \Rightarrow 0.333 \text{ m}$$

$$\text{Compute Maximum Doppler Shift (f}_d\text{)} = v/\lambda$$

$$f_d = \frac{30 \text{ m/s}}{\frac{1}{3} \text{ m}} \Rightarrow 30 \times 3 \Rightarrow 90 \text{ Hz}$$

10) Given

$$\text{Packet Size} = 1500 \text{ bytes} \Rightarrow 1500 \times 8 = 12,000 \text{ bits}$$

$$\text{Data Rate} = 54 \text{ Mbps} \Rightarrow 54 \times 10^6 \text{ bps} = 54,000,000 \text{ bps}$$

Transmission Time formula

$$T_e = \frac{\text{Packet Size (in bits)}}{\text{Transmission Rate (Mbps)}} \Rightarrow \frac{12,000}{54,000,000}$$

$$= \frac{12}{54,000} = \frac{1}{4500} \text{ sec} \approx 0.0002222$$

Unit Conversion

$$\text{in milliseconds (ms)} = 0.00022222 \times 1000 = 0.222 \text{ ms}$$

$$\text{in microseconds (us)} = 0.00022222 \times 1000000 \Rightarrow 222.22 \text{ us}$$